INTRODUCTION

Finding safe and affordable boarding homes in Cabanatuan City has been hard for professionals, students, and temporary workers for a long time. When people rely on unstructured sources like social media posts, word-of-mouth recommendations, or manual site visits, they often acquire the wrong information, see listings that are no longer available, and waste time. Most digital housing platforms nowadays, like Airbnb and Booking.com, are mostly for short-term rentals in big cities. This means that smaller cities and local boarding houses don't get enough attention. This imbalance makes it hard for property owners in rural areas to find tenants because they can't see them or use centralized services. Research indicates that short-term rental platforms are predominantly located in urban areas, so transforming housing markets and marginalizing conventional lodgings in smaller villages (Garay-Tamajón *et al.*, 2022). Airbnb's quick expansion has upset the accommodation industry in the Philippines. It has helped towns with a lot of tourists and hurt local boarding houses (Tumbali, 2020).

This study presents *Makikibahay:* A Web-Based Boarding House Finder with Smart Filtering and a 3D Room Preview for Cabanatuan City to address these issues. The system is planned as a full-stack, location-aware web app that brings together all boarding house listings, uses smart filtering, and lets users view 3D room previews. Through geolocation-based suggestions and onboarding questionnaires, the platform tailors search results to user type (worker or student), spending limit, and distance from locations such as offices or schools. This method helps people make better choices, cuts down on unnecessary travel, and gives property owners more control over their listings online. In the same way, Silva, Nogueira, and Bonaldo (2022) wrote about how location-aware algorithms might improve housing recommendations in "Location Matters: Leveraging Multi-Resolution Geo-Embeddings for Housing Search." These studies show how important location-based and user-centered housing solutions are becoming. This directly supports the idea behind platforms like Makikibahay, which want to make it easier to find boarding houses in smaller cities.

The importance of this study is that it could make the process of finding a home in Cabanatuan City more up-to-date. *Makikibahay* uses web technologies, geospatial computation, and user-centered design to make it easier for people to find a boarding house and build trust between renters and landlords. Pathan *et al.* (2023) highlighted the significance of location-based PG accommodation platforms in enhancing housing searches via geolocation and onboarding questions. The system also helps people go online by providing a mobile-first, easy-to-use interface that works on a wide range of devices. This project aims to find a scalable, ethical way to solve one of Cabanatuan's most persistent housing problems. It will also add to the larger conversation about how technology might help with housing problems in cities. It applies and is pertinent to various fields, including [1] Information Technology, showcasing the practical use of full-stack web development, GIS algorithms, and 3D visualization in addressing real-world challenges. [2] Knowledge of localized housing solutions and the contribution of digital platforms to enhancing accessibility can be gained from urban studies and housing management. [3] Business and Entrepreneurship gives property owners current tools to make their properties more

competitive and keep tenants interested. [4] Human-centered design emphasizes the significance of customization, accessibility, and user experience in system development.

Objectives of the Study

In Cabanatuan City, the search for reliable, affordable boarding houses remains a challenge for students, professionals, and workers. Traditional methods such as word-of-mouth referrals, social media posts, or manual site visits often result in outdated information, wasted time, and limited accessibility. Property owners likewise struggle to reach potential tenants effectively, as most existing platforms are designed for larger metropolitan areas or short-term rentals. These issues highlight the need for a centralized, transparent, and user-friendly system that modernizes the housing search process while empowering property owners with digital tools.

The study, therefore, aims to develop *Makikibahay: A Web-Based Boarding House Finder with Smart Filtering and 3D Room Preview for Cabanatuan City*. By integrating web technologies, geospatial computation, and user-centered design, the system seeks to improve efficiency, transparency, and trust in the boarding house rental market.

Specifically, the study seeks to:

- **Centralize boarding house information** in Cabanatuan City by creating a structured database of verified listings, complete with photos, pricing, amenities, and rules.
- **Improve reliability and transparency** in the housing search process through verified property details and user reviews, reducing misinformation and outdated listings.
- **Reduce unnecessary travel** by integrating immersive 3D room previews that allow tenants to virtually explore rooms before visiting.
- **Accelerate decision-making** using smart filtering and hybrid recommendation systems based on user profiles, budgets, and proximity to schools or workplaces.
- **Empower property owners** with digital tools such as dashboards for listing management, room availability tracking, and tenant communication.
- **Promote digital inclusion** by designing a mobile-first, accessible interface adaptable to various devices, ensuring usability for all types of users.
- **Establish a scalable and ethical solution** that can be adapted to other urban areas facing similar housing challenges, contributing to the modernization of local rental markets.

Significance of the study

The development of *Makikibahay: A Web-Based Boarding House Finder with Smart Filtering and 3D Room Preview for Cabanatuan City* is significant because it addresses one of the most persistent challenges in the local housing market: the lack of reliable, centralized, and transparent information about boarding houses. By integrating web technologies, geospatial computation, and user-centered design, the system modernizes the housing search process and creates meaningful benefits for multiple stakeholders.

For Students and Professionals

The technology makes it easy and dependable to choose boarding houses that fit their budget, location choices, and lifestyle needs. Tenants can make sensible choices without wasting time and money on unnecessary site visits thanks to clever filtering and 3D room previews that let them see how the room will look. Lovitos (2019) studied the housing preferences of college students in the Philippines, emphasizing that accessibility, cost, and quality of living circumstances are essential considerations in housing choices.

For Property Owners

Makikibahay empowers landlords with digital tools for listing management, availability tracking, and tenant communication. This increases their visibility in the local rental market, reduces reliance on informal advertising, and fosters trust through verified listings and user reviews. Leona et al. (2023) designed AppStay, a web-based reservation system for Cabanatuan City. However, it was mostly for booking and didn't have any tools to help landlords, including managing listings and talking to tenants.

For the Local Community

The project contributes to digital inclusion by offering a mobile-first, accessible interface that can be used across different devices. It promotes transparency and efficiency in the rental process, strengthening trust between tenants and landlords while supporting the modernization of Cabanatuan City’s housing sector. Accessible, mobile-first platforms are essential as mobile devices frequently serve as the primary means of internet access in developing countries (Selwyn, 2020; Robinson et al., 2015).

For the Academic and Professional Fields

The study generates knowledge on the integration of full-stack web development, GIS algorithms, and immersive visualization in solving localized housing challenges. It demonstrates how information technology can be applied to urban housing management, business innovation, and human-centered design. Furthermore, the system serves as a scalable model that can be adapted to other urban areas facing similar housing issues. Also, immersive visualization methods like 3D modeling and virtual reality have been found to improve urban planning and housing design by giving people who are involved in the process a realistic and interactive look at living places (Portman, Natapov, & Fisher-Gewirtzman, 2015).